Gabriel's Horn: A Revolutionary Tale

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Gabriel's Horn is the surface generated when the graph of the function $f(x) = x^{-1}$, defined for $x \geq 1$, is revolved around the x-axis as in FIGURE 1. The Horn was discovered in 1641 by Evangelista Torricelli, who established its celebrated properties, infinite surface area and finite volume—which we will refer to together as the *Horn Property*. The proper name Gabriel refers to the Archangel Gabriel, who in some religious traditions is viewed as the messenger of God, heralding the End of Days with a trumpet blast. Coupling the divine with the infinite completes the metaphor for the Horn, which is elsewhere called *Torricelli's trumpet* or the *infinite paint can*. The latter description gives rise to the "painter's paradox": How can a paint can full of paint contain insufficient liquid to coat its interior?

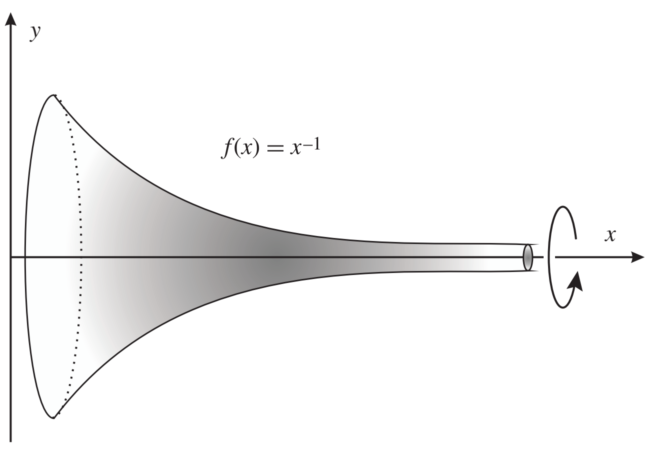

Figure 1 Gabriel's Horn

When first announced, the paradoxical nature of the Horn caused a furor, evoking spirited philosophical discussions on the nature of the infinite and leading the English philosopher Thomas Hobbes to exclaim that believing Torricelli was madness: "…to understand this for sense, it is not required that a man should be a geometrician or a logician, but that he should be mad" [**9**].

The novelty of the Horn spawned other seemingly fantastic examples, such as the drinking vessel of Huygens and deSluze [**10**], and even foreshadowed some of the curious properties of fractal sets such as *Koch's snowflake*, which encloses a finite area with an infinite perimeter [**17**].

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Torricelli's own words express his wonder at the example, clearly illuminating the difficulty that mathematicians of his day had with infinitesimals and quadrature, concepts which had kept no less a mind than Archimedes' from fully discovering the calculus a millennia earlier.

> It may seem incredible that although this solid has an infinite length, nevertheless none of the cylindrical surfaces we considered has an infinite length but all of them are finite.
>
> *Evangelista Torricelli*

Here, "cylindrical surfaces" refers to nested right circular cylinders whose axes are the x-axis and that are inscribed inside the Horn. Each of these cylinders has the same finite surface area and so too, it would seem, must their degenerate limit. However, the degenerate limit is the infinitely long x-axis. Without the language of calculus, Torricelli was unable to resolve this apparent paradox [**12**]. A more thorough explanation of Torricelli's approach can be found in a recent article in this MAGAZINE [**2**].

The paradox of the infinite paint can disappears when we realize that it is the difference in dimensions that creates the illusion of impossibility. We cannot paint the two-dimensional Horn with a uniformly thick coating of three-dimensional paint. For the Horn, we would have to reduce the thickness as we paint to the right. If we are careful, and reduce the thickness quickly enough, any amount of paint will do: We could use an eyedropper full of paint to coat the can.

Given the language of calculus, it is easy to establish the Horn's volume V and surface area A:

$$V = \int_1^\infty \pi [f(x)]^2\, dx = \pi \int_1^\infty \frac{1}{x^2}\, dx = \pi,$$

and

$$\begin{aligned} A &= \int_1^\infty 2\pi f(x)\sqrt{1+[f'(x)]^2}\, dx \\ &= 2\pi \int_1^\infty \frac{\sqrt{1+\frac{1}{x^4}}}{x}\, dx > 2\pi \int_1^\infty \frac{1}{x}\, dx = \infty. \end{aligned}$$

Evidently, the critical issue is one of convergence. The phenomenon exhibited by Gabriel's Horn is closely related to the convergence of certain p-series. We note that the defining curve, $f(x) = x^{-1}$, is one among many that will generate an object with the finite volume, infinite surface area property. For example, the profile function $g(x) = x^{-p}$, for any p satisfying $\frac{1}{2} < p \leq 1$, will also work. As an interesting aside, Fleron [**8**] uses step functions to create a discrete version of Gabriel's Horn—*Gabriel's wedding cake*—which we "can eat but cannot frost" (FIGURE 2).

Despite the age and celebrity of Gabriel's Horn, we find no evidence that higher-dimensional analogues were investigated before the last decade. In a 2004 article [**7**] in this MAGAZINE, Eisenberg provides an intuitive introduction to 3-dimensional hypersurfaces of revolution in 4-dimensional Euclidean space, including both heuristic arguments and analytic proofs of the associated volume formulas. He notes that Gabriel's Horn generalizes neatly to this setting by finding 3-dimensional hypersurfaces of revolution with the Horn Property—infinite "surface area" (properly, 3-volume) enclosing a region of finite 4-volume. Three years later, Aberra and Agrawal [**1**] extended these results to arbitrary dimensions $n \geq 2$ by presenting the volume formulas associated with an n-dimensional hypersurface of revolution. There, the evident pattern

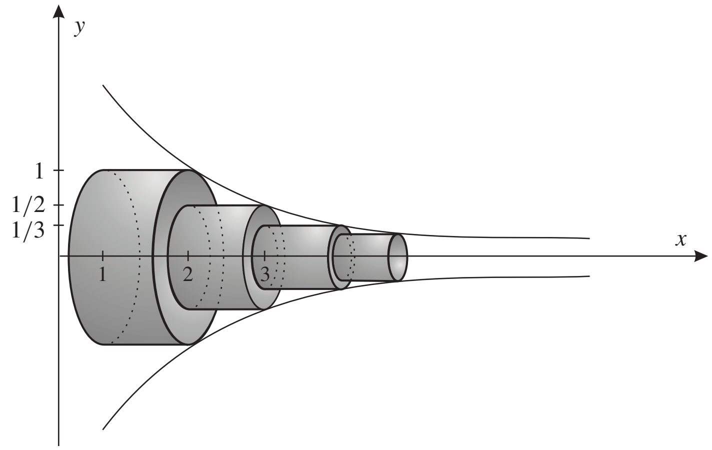

Figure 2 Gabriel's wedding cake

of Gabriel's Horn for higher-dimensional rotational hypersurfaces is briefly discussed (see also Example 3 later in this article).

Here, we introduce more general objects, called *spherical arrays*, that naturally generalize surfaces of revolution and contain hypersurfaces of revolution as a proper subclass. The rich symmetry of these objects can be exploited to develop their volume formulas with relative ease. It is our hope that the explicit nature of these calculations will stimulate interest in differential geometry for those students who have not seen calculus in higher-dimensional Euclidean spaces. An immediate consequence of our study is the development of new examples of hypersurfaces that exhibit properties reminiscent of those for which Gabriel's Horn became a *cause célèbre*.

Generalizing surfaces of revolution

A *surface of revolution*, as typically studied in the first year of a calculus course, is a surface H in $\mathbb{R}^3$ such that the cross-sections orthogonal to a particular line, which we label the x-axis and refer to as the *axis of rotation*, are circles centered on the x-axis, as illustrated in FIGURE 3. There, the notation $S^1(f(x))$ refers to a cross-sectional circle of radius $f(x)$, where $f : \Omega \to [0, \infty)$ is a C^1 function, which we call the *profile function* of H, and $\Omega \subset \mathbb{R}$ is the domain of f.

Algebraically, we write $H = \left\{(x_1, x_2, x) \in \mathbb{R}^2 \times \Omega \subset \mathbb{R}^3 : x_1^2 + x_2^2 = [f(x)]^2\right\}$. Note that if we fix a specific $x = x_0$, the equation reduces to $x_1^2 + x_2^2 = [f(x_0)]^2$, which represents a circle centered about the x-axis at x_0, with radius $f(x_0)$. Intuitively, we may visualize H as a collection of circles stacked alongside one another, with radii varying smoothly according to the value of the profile function. We stress that in this definition, we are decomposing $\mathbb{R}^3$ into two parts: one domainal direction and two rotational directions.

A natural abstraction to 4-dimensional space only requires replacing the cross-sectional circles, which are just 1-dimensional spheres, with cross-sectional 2-dimensional spheres. In particular, we may define a *hypersurface of revolution* H in $\mathbb{R}^4$ algebraically as $H = \left\{(x_1, x_2, x_3, x) \in \mathbb{R}^3 \times \Omega \subset \mathbb{R}^4 : x_1^2 + x_2^2 + x_3^2 = [f(x)]^2\right\}$, where f represents the profile function of H. The intuitive definition is unchanged: A hypersurface of revolution may be thought of as a collection of spheres, stacked alongside one another, with radius varying according to the value of the profile function. And as above, this definition requires a decomposition of $\mathbb{R}^4$ into one domainal dimension and three rotational directions.

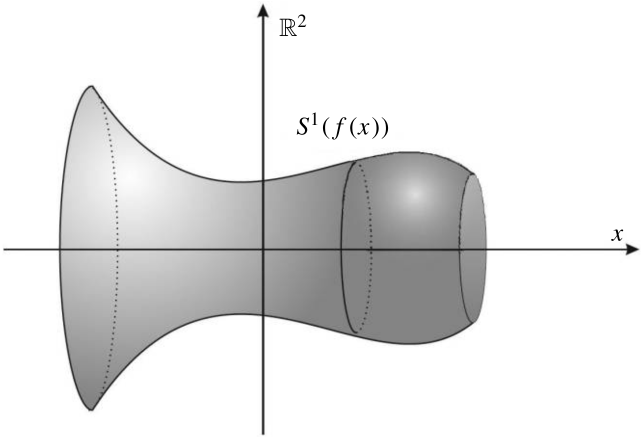

Figure 3 A surface of revolution

But suppose we were to decompose $\mathbb{R}^4$ differently? For example, we may decide to dedicate two dimensions of the ambient $\mathbb{R}^4$ for the domain of the profile function, reserving the remaining two dimensions for rotation. This is formalized in the following definition.

DEFINITION. Given a domain $\Omega \subset \mathbb{R}^2$ and a C^1 profile function $f : \Omega \to [0, \infty)$, we define the *spherical array* $\mathcal{S}_2^3$ *generated by* f to be

$$\mathcal{S}_2^3 = \left\{(x_1, x_2, y_1, y_2) \in \mathbb{R}^2 \times \Omega : x_1^2 + x_2^2 = [f(y_1, y_2)]^2\right\}, \tag{1}$$

where the subscript represents the number of rotational dimensions and the superscript represents the dimension of the spherical array. Note that a surface of revolution would then be an $\mathcal{S}_2^2$ spherical array.

As above, equation (1) is the algebraic statement that for each fixed $y = (y_1, y_2) \in \Omega$, a cross-section of $\mathcal{S}_2^3$ taken perpendicular to Ω is a circle of radius $f(y_1, y_2)$.

Considering all profile functions of two variables would be an overbroad investigation, so we focus on those functions dependent on the norm of $y = (y_1, y_2) \in \Omega \subset \mathbb{R}^2$. Geometrically, the level sets of any such function are (unions of) circles centered at the origin of $\mathbb{R}^2$, and so it is natural to assume that the domain Ω is a ball centered at the origin. We refer to the resulting spherical arrays as *radially symmetric*. In addition to the rotational symmetry exhibited by all spherical arrays, observe that radially symmetric spherical arrays maintain additional symmetry in the domainal directions. See FIGURE 4, where the $y_1 y_2$-plane represents the domainal $\mathbb{R}^2$, and the vertical axis represents the rotational $\mathbb{R}^2$. Note that in FIGURE 4, the point labeled $S^1(f(\|y\|))$ on $\mathcal{S}_2^3$ is a circle.

We remark that, mathematically, spherical arrays appear in a number of guises. If the profile function does not take on the value 0, a spherical array may be viewed topologically as a trivial sphere bundle with base space Ω. Geometrically, such an array has the additional structure of a warped product. Finally, radially symmetric spherical arrays can be studied from the perspective of their isometry groups, which necessarily contain a copy of $O(2) \times O(2)$, where $O(m)$ represents the group of $m \times m$ orthogonal matrices. The first copy of $O(2)$ arises from the two rotational dimensions, while the second copy of $O(2)$ corresponds to the two domainal directions.

In what follows, we assume that all spherical arrays are radially symmetric. In particular, the notation $\mathcal{S}_2^3$ always represents a radially symmetric spherical array.

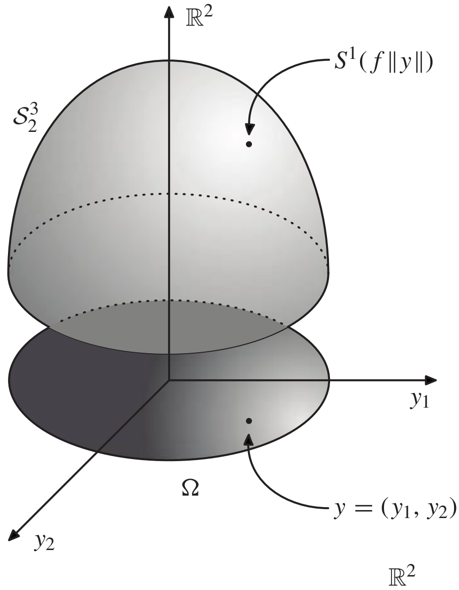

Figure 4 A radially symmetric spherical array in $\mathbb{R}^4$

Volume formula for $\mathcal{S}_2^3$

In the final weeks of a multivariable calculus course, students are required to perform calculations involving parametrized curves and surfaces. The techniques for computing the arc length of a parametrized curve or the surface area of a parametrized surface follow the same recipe: After finding a suitable parametrization, the arc length (1-volume) or surface area (2-volume) is found by integrating some quantity, dependent in a natural way on the parametrization and its first partial derivatives, over the domain of the parametrization. This is illustrated in the following example.

EXAMPLE 1. If $\gamma : [a, b] \to \mathbb{R}^3 : t \mapsto (x(t), y(t), z(t))$ is a parametrized space curve, the *arc length* of $\gamma(t)$ is $\int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\, dt$.

Similar techniques can be used to compute the k-volume of any smooth k-dimensional object H in $\mathbb{R}^{n+1}$. Let σ be a parametrization of H and $\sigma_1, \dots, \sigma_k$ represent the partial derivatives of σ with respect to the k coordinate directions. Then each σ_i is a vector in $\mathbb{R}^{n+1}$, and we can use these vectors to populate a $k \times k$ matrix as follows:

$$g = \begin{pmatrix} \langle \sigma_1, \sigma_1 \rangle & \langle \sigma_1, \sigma_2 \rangle & \cdots & \langle \sigma_1, \sigma_k \rangle \\ \langle \sigma_2, \sigma_1 \rangle & \langle \sigma_2, \sigma_2 \rangle & \cdots & \langle \sigma_2, \sigma_k \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \sigma_k, \sigma_1 \rangle & \langle \sigma_k, \sigma_2 \rangle & \cdots & \langle \sigma_k, \sigma_k \rangle \end{pmatrix}, \tag{2}$$

where the notation $\langle \cdot, \cdot \rangle$ represents the dot product in $\mathbb{R}^{n+1}$. The k-volume of H is then given by integrating $\sqrt{|\det(g)|}$ over the domain of the parametrization.

The quantity $\sqrt{|\det(g)|}$ has a nice geometric interpretation. Given a collection of vectors $v_1, \dots, v_m$ in $\mathbb{R}^{n+1}$, the matrix formed by taking pairwise dot products of these vectors, as in (2), is called the *Gram matrix* (or *Gramian*) of $v_1, \dots, v_m$. Then, $\sqrt{|\det(g)|}$ represents the m-volume of the m-dimensional parallelepiped spanned by the vectors $v_1, \dots, v_m$. Intuitively, the volume of an object H parametrized by σ is given by summing, over the domain of the parametrization, the volumes of the infinitesimal parallelepipeds formed by the partial derivatives of σ.

In differential geometry, the matrix in (2) is called the *first fundamental form* and appears in all distance and volume related calculations. The reader may wish to consult [**3**] or [**13**] for more information.

EXAMPLE 2. Given a vector $v = (a, b) \in \mathbb{R}^2$, the Gramian of v is the 1×1 matrix with entry $a^2 + b^2$, so the 1-volume of the parallelepiped (the length of the vector v) is given by $\sqrt{|\det(g)|} = \sqrt{a^2 + b^2}$, which is just the statement of the Pythagorean Theorem! Repeating for two vectors v_1, v_2 in $\mathbb{R}^2$ gives the area of the parallelogram spanned by v_1 and v_2. This extends to a set of vectors $v_1, \ldots, v_m$ in $\mathbb{R}^n$ to yield m-volume in n-dimensional space. See the excellent article by Porter [**14**] for more detail on this generalization of the Pythagorean Theorem.

EXERCISE. The procedure described above can be used to verify the formulas for arc length and surface area as presented in a multivariable calculus course. That is, the formula in Example 1 may be derived by computing the quantity $\sqrt{|\det(g)|}$, where $g = (\,\langle \gamma'(t), \gamma'(t)\rangle\,)$ is the Gram matrix induced by the parametrization γ. Next, given a subset $U \subset \mathbb{R}^2$, let $\sigma : U \to \mathbb{R}^3 : (u, v) \mapsto \sigma(u, v)$ be a smooth parametrization of a surface S. Then the surface area of S is given by the formula $A(S) = \int_U |\sigma_u \times \sigma_v|\, dU$ (see [**16**], p. 1076). Check that if

$$g = \begin{pmatrix} \langle \sigma_u, \sigma_u \rangle & \langle \sigma_u, \sigma_v \rangle \\ \langle \sigma_v, \sigma_u \rangle & \langle \sigma_v, \sigma_v \rangle \end{pmatrix}$$

is the Gram matrix of the parametrization σ, then $\sqrt{|\det(g)|}$ agrees with the magnitude of the cross product of σ_u and σ_v.

The volume of a radially symmetric spherical array $\mathcal{S}_2^3$ may be computed according to the following three-step process:

1. Find a parametrization σ of $\mathcal{S}_2^3$,
2. Compute the matrix g corresponding to σ,
3. Integrate the quantity $\sqrt{|\det(g)|}$ over the domain of σ.

Theoretically, this three-step program may be used to compute the k-volume of any smooth k-dimensional object in $\mathbb{R}^{n+1}$. In practice, the most common obstacle to carrying out this program occurs at the first step, since finding a parametrization of the target object may be difficult. However, the rich symmetry of a radially symmetric spherical array can be used to find a suitable parametrization. In particular, two sets of polar coordinates can be used: one set for the domain Ω (since it is assumed to be a round ball), and one set to parametrize the cross-sectional circles. If Ω is a closed ball of radius $R \in (0, \infty]$, (where $R = \infty$ corresponds to the case $\Omega = \mathbb{R}^2$), then a parametrization σ of $\mathcal{S}_2^3$ is given by

$$\sigma(\phi, r, \theta) = (f(r)\cos\phi,\ f(r)\sin\phi,\ r\cos\theta,\ r\sin\theta), \qquad (3)$$
$$\text{where } 0 \le r \le R \text{ and } 0 \le \phi, \theta \le 2\pi.$$

Indeed, one may check that (3) is equivalent to (1). Note that the final two coordinates, r and θ, parametrize the ball Ω, and for fixed r and θ, the first coordinate, ϕ, parametrizes the cross-sectional circles of radius $f(r)$.

The second step is to compute the matrix g. We first compute the partial derivatives

$$\begin{aligned}\sigma_\phi &= (-f(r)\sin\phi,\ f(r)\cos\phi,\ 0,\ 0),\\ \sigma_r &= (f'(r)\cos\phi,\ f'(r)\sin\phi,\ \cos\theta,\ \sin\theta),\\ \sigma_\theta &= (0,\ 0,\ -r\sin\theta,\ r\cos\theta).\end{aligned}$$

So, we have

$$g = \begin{pmatrix} \langle\sigma_\phi,\sigma_\phi\rangle & \langle\sigma_\phi,\sigma_r\rangle & \langle\sigma_\phi,\sigma_\theta\rangle \\ \langle\sigma_r,\sigma_\phi\rangle & \langle\sigma_r,\sigma_r\rangle & \langle\sigma_r,\sigma_\theta\rangle \\ \langle\sigma_\theta,\sigma_\phi\rangle & \langle\sigma_\theta,\sigma_r\rangle & \langle\sigma_\theta,\sigma_\theta\rangle \end{pmatrix} = \begin{pmatrix} [f(r)]^2 & 0 & 0 \\ 0 & 1+[f'(r)]^2 & 0 \\ 0 & 0 & r^2 \end{pmatrix}.$$

We are left with the integration of the quantity $\sqrt{|\det(g)|}$ over the domain of the parametrization.

$$\begin{aligned}\mathrm{Vol}(\mathcal{S}_2^3) &= \int_0^{2\pi}\int_0^R\int_0^{2\pi} \sqrt{|\det(g)|}\,d\phi\,dr\,d\theta\\ &= \int_0^{2\pi}\int_0^R\int_0^{2\pi} r\cdot f(r)\cdot\sqrt{1+[f'(r)]^2}\,d\phi\,dr\,d\theta\\ &= 4\pi^2\int_0^R r\cdot f(r)\cdot\sqrt{1+[f'(r)]^2}\,dr. \qquad (4)\end{aligned}$$

This completes the three-step process, and we are rewarded with the volume of a radially symmetric spherical array.

Volume formula for $\mathcal{E}_2^4$

This process can be replicated to find the volume of the solid $\mathcal{E}_2^4$ enclosed by $\mathcal{S}_2^3$. We formally define this object by

$$\mathcal{E}_2^4 = \left\{(x_1, x_2, y_1, y_2) \in \mathbb{R}^2\times\Omega : x_1^2 + x_2^2 \le [f(y_1, y_2)]^2\right\}.$$

Assuming again that f is radially symmetric, we can parametrize $\mathcal{E}_2^4$ by modifying the parametrization of $\mathcal{S}_2^3$, in the sense that we need only change the first two coordinates so that they parametrize cross-sectional solid disks of radius $f(r)$, instead of circles of radius $f(r)$. Thus, a parametrization σ of $\mathcal{E}_2^4$ is given by

$$\sigma(\rho,\phi,r,\theta) = (\rho\cos\phi,\ \rho\sin\phi,\ r\cos\theta,\ r\sin\theta),$$
$$\text{where } 0\le\rho\le f(r),\ 0\le r\le R,\ \text{and } 0\le\phi,\theta\le 2\pi.$$

We then have

$$\begin{aligned}\sigma_\rho &= (\cos\phi, \sin\phi, 0, 0),\\ \sigma_\phi &= (-\rho\sin\phi, \rho\cos\phi, 0, 0),\\ \sigma_r &= (0, 0, \cos\theta, \sin\theta),\\ \sigma_\theta &= (0, 0, -r\sin\theta, r\cos\theta),\end{aligned}$$

so that

$$g = \begin{pmatrix} \langle\sigma_\rho, \sigma_\rho\rangle & \langle\sigma_\rho, \sigma_\phi\rangle & \langle\sigma_\rho, \sigma_r\rangle & \langle\sigma_\rho, \sigma_\theta\rangle \\ \langle\sigma_\phi, \sigma_\rho\rangle & \langle\sigma_\phi, \sigma_\phi\rangle & \langle\sigma_\phi, \sigma_r\rangle & \langle\sigma_\phi, \sigma_\theta\rangle \\ \langle\sigma_r, \sigma_\rho\rangle & \langle\sigma_r, \sigma_\phi\rangle & \langle\sigma_r, \sigma_r\rangle & \langle\sigma_r, \sigma_\theta\rangle \\ \langle\sigma_\theta, \sigma_\rho\rangle & \langle\sigma_\theta, \sigma_\phi\rangle & \langle\sigma_\theta, \sigma_r\rangle & \langle\sigma_\theta, \sigma_\theta\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \rho^2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & r^2 \end{pmatrix}.$$

It follows that the volume of $\mathcal{E}_2^4$ is given by

$$\begin{aligned} \mathrm{Vol}(\mathcal{E}_2^4) &= \int_0^{2\pi} \int_0^R \int_0^{2\pi} \int_0^{f(r)} \sqrt{|\det(g)|}\, d\rho\, d\phi\, dr\, d\theta \\ &= \int_0^{2\pi} \int_0^R \int_0^{2\pi} \int_0^{f(r)} \rho \cdot r\, d\rho\, d\phi\, dr\, d\theta \\ &= 2\pi^2 \int_0^R r \cdot [f(r)]^2\, dr. \end{aligned} \tag{5}$$

Volume formulas for $\mathcal{S}_k^n$

The spherical arrays discussed in the previous section arise from a specific decomposition of $\mathbb{R}^4$, but the setup generalizes neatly to higher dimensions and other decompositions of the dimension of the ambient space. Given $n \geq 2$, choose an integer k $(1 < k \leq n)$ to represent the number of rotational dimensions. Now, given $\Omega \subset \mathbb{R}^{n-k+1}$ and a profile function $f : \Omega \to [0, \infty)$, define a spherical array $\mathcal{S}_k^n$ implicitly as

$$\begin{aligned} \mathcal{S}_k^n = \big\{ &(x_1, \dots, x_k, y_1, \dots, y_{n-k+1}) \in \mathbb{R}^k \times \Omega \subset \mathbb{R}^{n+1} : \\ &x_1^2 + \cdots + x_k^2 = [f(y_1, \dots, y_{n-k+1})]^2 \big\}. \end{aligned}$$

The spherical arrays $\mathcal{S}_2^3$ discussed above occur when $n = 3$ and $k = 2$. The special case $k = n$ yields an n-dimensional hypersurface of revolution.

We proceed with a heuristic argument for the volume formulas of a generic, radially symmetric spherical array. Let $a_{m-1}(\delta)$ represent the $(m-1)$-volume of the $(m-1)$-dimensional sphere of radius δ and $v_m(\delta)$ represent the m-volume of the m-dimensional solid ball of radius δ. The exact values of $a_{m-1}(\delta)$ and $v_m(\delta)$ are not needed, but we do make use of the fact that there exists a positive constant C_m such that

$$a_{m-1}(\delta) = mC_m\delta^{m-1} \qquad \text{and} \qquad v_m(\delta) = C_m\delta^m. \tag{6}$$

When $m = 2$, $a_1(\delta) = 2\pi\delta$ is the circumference of a circle of radius δ, and $v_2(\delta) = \pi\delta^2$ represents the area of a disk of radius δ. Therefore, we can rewrite (4) and (5) as

$$\begin{aligned} \mathrm{Vol}(\mathcal{S}_2^3) &= \int_0^R 2\pi r \cdot 2\pi f(r) \cdot \sqrt{1 + [f'(r)]^2}\, dr \\ &= \int_0^R a_1(r) \cdot a_1(f(r)) \cdot \sqrt{1 + [f'(r)]^2}\, dr, \quad \text{and} \\ \mathrm{Vol}(\mathcal{E}_2^4) &= \int_0^R 2\pi r \cdot \pi [f(r)]^2\, dr \\ &= \int_0^R a_1(r) \cdot v_2(f(r))\, dr. \end{aligned}$$

The formulas above generalize neatly to higher-dimensional spherical arrays.

THEOREM 1. *Given $n \geq 2$ and $1 < k \leq n$, let Ω be the $(n-k+1)$-dimensional solid ball of radius R. If $f : \Omega \to [0, \infty)$ is the profile function of a radially symmetric spherical array $\mathcal{S}_k^n$, then the volume of $\mathcal{S}_k^n$ and its enclosed solid $\mathcal{E}_k^{n+1}$ are, respectively, given by*

$$\begin{aligned}\operatorname{Vol}(\mathcal{S}_k^n) &= \int_0^R a_{n-k}(r) \cdot a_{k-1}(f(r)) \cdot \sqrt{1 + [f'(r)]^2}\, dr \\ &= a_{n-k}(1) \cdot a_{k-1}(1) \int_0^R r^{n-k} \cdot [f(r)]^{k-1} \cdot \sqrt{1 + [f'(r)]^2}\, dr, \qquad (7)\\ \operatorname{Vol}(\mathcal{E}_k^{n+1}) &= \int_0^R a_{n-k}(r) \cdot v_k(f(r))\, dr \\ &= a_{n-k}(1) \cdot v_k(1) \int_0^R r^{n-k} \cdot [f(r)]^k\, dr. \qquad (8)\end{aligned}$$

Observe that in each case, the second equality follows from (6). For the first equalities, the same technique as above can be used. When $n = 5$ and $k = 3$, a radially symmetric spherical array can be parametrized with two sets of spherical coordinates as follows:

$$\begin{aligned}\sigma(\phi_1, \phi_2, r, \theta_1, \theta_2) = (&f(r) \sin\phi_1 \sin\phi_2,\ f(r)\sin\phi_1\cos\phi_2,\ f(r)\cos\phi_1,\\ &r\sin\theta_1\sin\theta_2, r\sin\theta_1\cos\theta_2, r\cos\theta_1),\\ \text{where } 0 \leq r \leq R,\ & 0 \leq \phi_1, \theta_1 \leq \pi,\ 0 \leq \phi_2, \theta_2 \leq 2\pi.\end{aligned}$$

One may check, after computing the partial derivatives, that the matrix g is given by

$$g = \begin{pmatrix} [f(r)]^2 & 0 & 0 & 0 & 0 \\ 0 & \sin^2\phi_1\,[f(r)]^2 & 0 & 0 & 0 \\ 0 & 0 & 1 + [f'(r)]^2 & 0 & 0 \\ 0 & 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & 0 & \sin^2\theta_1\, r^2 \end{pmatrix}.$$

The formula for $\operatorname{Vol}(\mathcal{S}_3^5)$ now follows from setting up the volume integral and applying equation (1.10) from **[1]**. The volume of the solid array may be computed similarly. In higher dimensions, two sets of *hyperspherical coordinates* [**18**] may be used.

Generalizing Gabriel's Horn

We now turn our attention to building n-dimensional spherical arrays with the Horn Property. Armed with equations (7) and (8), and the knowledge that $\int_1^\infty r^{-p}\, dr$ converges for $p > 1$, this is a straightforward task.

THEOREM 2. *Let $\Omega = \left\{y = (y_1, \ldots, y_{n-k+1}) \in \mathbb{R}^{n-k+1} : 1 \leq \|y\| = r\right\}$ and let $\mathcal{G}_k^n$ be the radially symmetric spherical array defined by*

$$\mathcal{G}_k^n = \Big\{(x_1, \ldots, x_k, y_1, \ldots, y_{n-k+1}) \in \mathbb{R}^k \times \Omega \subset \mathbb{R}^{n+1} :\\ x_1^2 + \cdots + x_k^2 = \left(y_1^2 + \cdots + y_{n-k+1}^2\right)^{-p}\Big\}.$$

Then $\mathcal{G}_k^n$ exhibits the Horn Property when $\frac{n-k+1}{k} < p \leq \frac{n-k+1}{k-1}$.

Proof. The hypersurface $\mathcal{G}_k^n$ is the radially symmetric spherical array generated by the profile function $f(r) = r^{-p}$ on the domain Ω. From (8), the volume enclosed by $\mathcal{G}_k^n$ is proportional to

$$\int_1^\infty r^{n-k} \cdot [f(r)]^k \, dr = \int_1^\infty r^{n-k} \cdot r^{-pk} \, dr = \int_1^\infty r^{n-k-pk} \, dr,$$

which is finite when $p > \frac{n-k+1}{k}$. On the other hand, we have

$$\begin{aligned}\int_1^\infty r^{n-k} \cdot [f(r)]^{k-1} \cdot \sqrt{1 + [f'(r)]^2} \, dr &\geq \int_1^\infty r^{n-k} \cdot [f(r)]^{k-1} \, dr \\ &= \int_1^\infty r^{n-k} \cdot r^{-p(k-1)} \, dr \\ &= \int_1^\infty r^{n-k-p(k-1)} \, dr,\end{aligned}$$

so by (7), the volume of $\mathcal{G}_k^n$ is infinite when $p \leq \frac{n-k+1}{k-1}$. ■

EXAMPLE 3. If $f(x) = x^{-p}$ is used to generate an n-dimensional hypersurface of revolution H in $\mathbb{R}^{n+1}$, then H has the Horn Property precisely when $\frac{1}{n} < p \leq \frac{1}{n-1}$ (see [**1**]).

More objects

So far, we have restricted our attention to hypersurfaces with spherical cross-sections. Can other objects be used as cross-sections? Certainly! In particular, nontrivial hypersurfaces arise when the cross-sectional objects are other than topological spheres. So long as the chosen object admits a reasonable parametrization, the volume formulas associated with the resulting hypersurface will be computable by the method outlined earlier. For example, a *torus* is the surface of revolution generated by revolving a circle about a coplanar, non-incident axis. If the circle has radius R_1 and the center of the circle is distance R_2 from the axis, the torus is said to have inner radius R_1 and outer radius R_2 (see FIGURE 5).

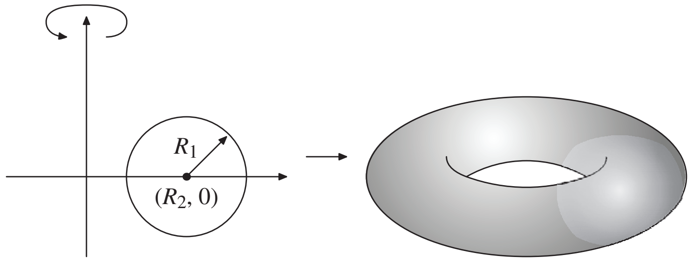

Figure 5 A torus as a surface of revolution in $\mathbb{R}^3$

Defining a hypersurface with toroidal cross-sections first requires a parametrization of the torus itself. Letting R_1 and R_2 be as above, the standard parametrization of a torus embedded in $\mathbb{R}^3$ is given by

$$\sigma(\phi_1, \phi_2) = (\cos\phi_1(R_2 + R_1\cos\phi_2),\ \sin\phi_1(R_2 + R_1\cos\phi_2),\ R_1\sin\phi_2),$$
$$\text{where } 0 \leq \phi_1, \phi_2 < 2\pi.$$

As before, assume that $\Omega \subset \mathbb{R}^2$ is a ball of radius R centered at the origin and that $f : \Omega \to (0, \infty)$ is a radially symmetric profile function. We then define a 4-dimensional *toroidal array* in $\mathbb{R}^5$ so that for each $y \in \Omega$, the cross-section in the three directions orthogonal to Ω is a torus with inner radius $f(r)$, $r = \|y\|$, and outer radius 4. In this case, to avoid self-intersection, we insist that $f(r) < 2$. Whence, a parametrization σ_1 for a toroidal array $\mathcal{T}$ is given by

$$\begin{aligned}\sigma_1(\phi_1, \phi_2, r, \theta) = (&\cos\phi_1(4 + f(r)\cos\phi_2),\ \sin\phi_1(4 + f(r)\cos\phi_2),\\ &f(r)\sin\phi_2,\ r\cos\theta,\ r\sin\theta),\\ \text{where } 0 \le r \le R &\text{ and } 0 \le \phi_1, \phi_2, \theta < 2\pi.\end{aligned}$$

As in the parametrization of a spherical array, note that the final two coordinates, r and θ, parametrize the ball Ω, and for fixed r and θ, the first two coordinates, ϕ_1 and ϕ_2, parametrize the cross-sectional torus with outer radius 4 and inner radius $f(r)$.

Similarly, a parametrization σ_2 for the solid T contained inside $\mathcal{T}$ is given by

$$\begin{aligned}\sigma_2(\rho, \phi_1, \phi_2, r, \theta) = (&\cos\phi_1(4 + \rho\cos\phi_2),\ \sin\phi_1(4 + \rho\cos\phi_2),\\ &\rho\sin\phi_2,\ r\cos\theta,\ r\sin\theta),\\ \text{where } 0 \le r \le R,\ 0 \le \rho &\le f(r), \text{ and } 0 \le \phi_1, \phi_2, \theta < 2\pi.\end{aligned}$$

EXERCISE. Show that the volume and enclosed volume of a radially symmetric toroidal array with outer radius 4 are given by

$$\begin{aligned}\mathrm{Vol}(\mathcal{T}) &= 32\pi^3 \int_0^R r \cdot f(r) \cdot \sqrt{1 + [f'(r)]^2}\, dr\\ &= 8\pi \cdot a_1(1) \cdot a_1(1) \int_0^R r \cdot f(r) \cdot \sqrt{1 + [f'(r)]^2}\, dr\\ &= 8\pi \cdot \mathrm{Vol}(\mathcal{S}_2^3),\end{aligned}$$

and

$$\begin{aligned}\mathrm{Vol}(T) &= 16\pi^3 \int_0^R r \cdot [f(r)]^2\, dr\\ &= 8\pi \cdot a_1(1) \cdot v_2(1) \int_0^R r \cdot f(r)^2\, dr\\ &= 8\pi \cdot \mathrm{Vol}(\mathcal{E}_2^4).\end{aligned}$$

It follows from the above formulas that if $\Omega = \left\{ y = (y_1, y_2) \in \mathbb{R}^2 : \|y\| \ge 1 \right\}$ with $f(r) = r^{-p}$ for $1 < p \le 2$, the resulting toroidal array has the Horn Property. Of course, the existence of topologically nontrivial versions of Gabriel's Horn is not surprising. For example, we may picture removing a smaller solid horn-shaped object from the interior of the ordinary, solid version of Gabriel's Horn. The cross-sections of the remaining object are annular disks centered along the axis of revolution, and the resulting 3-volume is decreased while the surface area of the object is increased—so the Horn Property is maintained. This type of construction is viable in any dimension. Explicitly, we may build such an object by generating a hypersolid of revolution from the region between the curves $g(x) = x^{-1}$ and $h(x) = x^{-2}$.

The interested reader is encouraged to consider an array with different cross-sections, or possibly a toroidal array where the outer radius $F(r)$ is not a constant. This collection of examples should illustrate that the novelty of new hypersurfaces with the Horn Property is limited only by the creativity of the reader.

Afterward

Gabriel's Horn is an unbounded surface of revolution with finite volume. In [**11**], a famous function from classical analysis is modified to develop a compact surface of revolution that has the Horn Property. However, the profile curve for this example is only differentiable and not of C^1 smoothness. Indeed, a compact surface of revolution (with profile function f) exhibiting the Horn Property cannot have C^1 smoothness, since this would imply that both f and f' are bounded, and so too, therefore, is the integral that defines the surface area. Similarly, since spherical arrays are dependent on a function of one variable, they cannot be both compact and C^1. However, spherical arrays that are both compact and smooth (in fact, analytic) have been studied in another context. In [**5**], the authors use these hypersurfaces to produce examples of objects that satisfy a higher-dimensional analogue of a classical property of the sphere whose discovery dates back to Archimedes—that the surface area of a zone of the sphere lying between two parallel planes depends only on the distance between the planes [**4, 6, 15**]. Also in [**5**], an alternate method for proving Theorem 1 is given.

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Summary We generalize hypersurfaces of revolution by allowing the profile curve to be dependent on more than one variable. We call these more general objects spherical arrays and we use them to introduce the reader to volume calculations, which are normally reserved for a more advanced course in differential geometry. The rich symmetry of the spherical arrays allow us to build objects with properties reminiscent of those for which Gabriel's Horn became a *cause célèbre*.

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Proof Without Words: An Electrical Proof of the AM-HM Inequality

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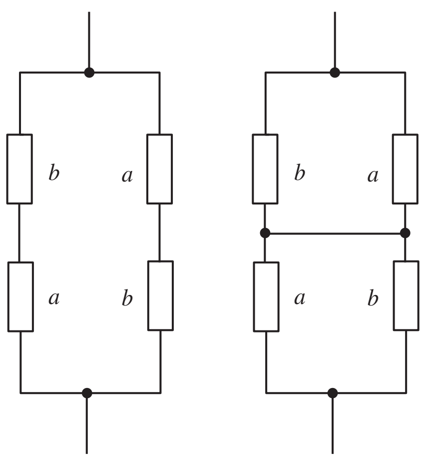

$$R_L = \frac{a+b}{2} \geq \frac{2ab}{a+b} = R_R$$

That is, AM $\geq$ HM; the arithmetic is at least equal to the harmonic mean. The inequality extends to the geometric mean as well: AM $\geq$ GM $\geq$ HM, because GM^2 = $AM \cdot HM$.

This proof has appeared in at least two books [**1, 2**], both of which explore the rich connections between mathematics and physics.

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Summary The AM-HM inequality arises from comparing the resistances in two circuits.

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